

AWS Three-Tier Architecture Project

AWS Three-Tier Student Application

1. Project Introduction

A three-tier architecture splits an application into three layers: a **presentation** tier (what the user sees), an **application** tier (business logic) and a **data** tier (database). This project builds a **Student Registration** web application on AWS in the Asia Pacific (Mumbai) region (ap-south-1). Students can be registered through a web form, and the saved records are shown in a Students List.

AWS services and software used: Amazon VPC (subnets, route tables, Internet Gateway, NAT Gateway), Amazon EC2 with security groups, Amazon RDS (MariaDB), EC2 Instance Connect, Nginx and Apache Tomcat 9.0.120.

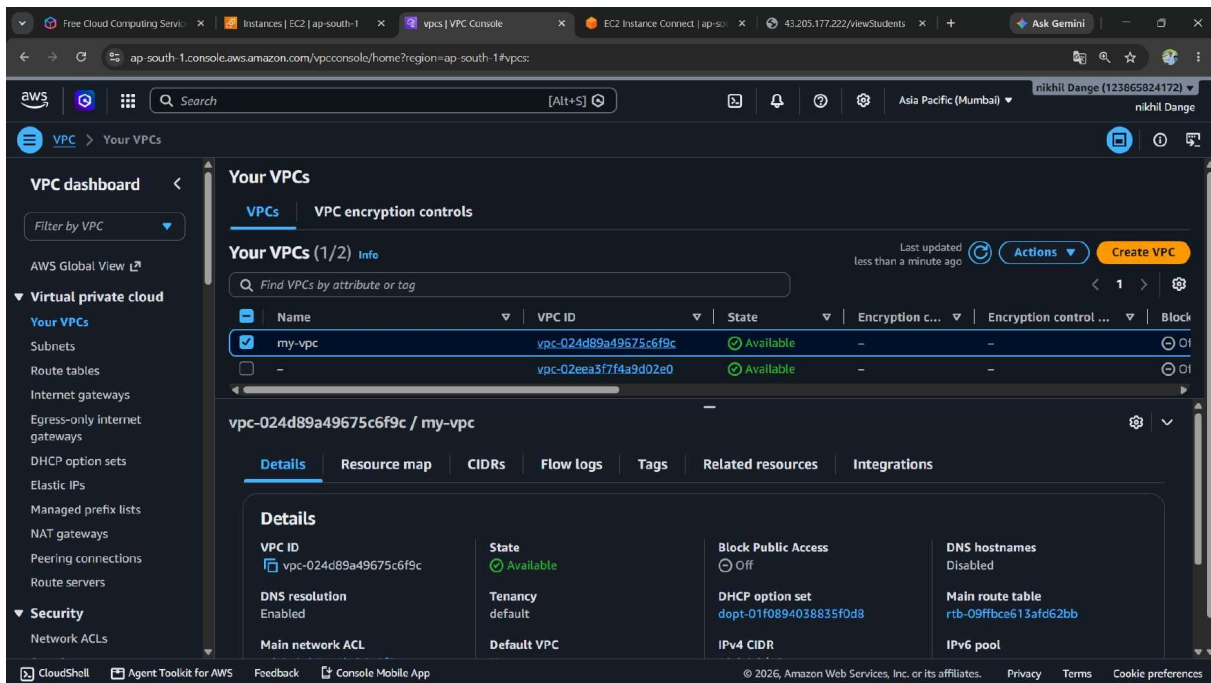
Part A — Network Setup (VPC)

Step 1 — VPC Foundation

What it shows: The VPC dashboard with the project VPC selected.

Details visible in the screenshot:

- Name: **my-vpc**; VPC ID: vpc-024d89a49675c6f9c
- State: Available; Tenancy: default
- DNS resolution: Enabled; DNS hostnames: Disabled
- Main route table: rtb-09ffbce613afd62bb
- IPv4 CIDR: Not visible in the provided screenshot

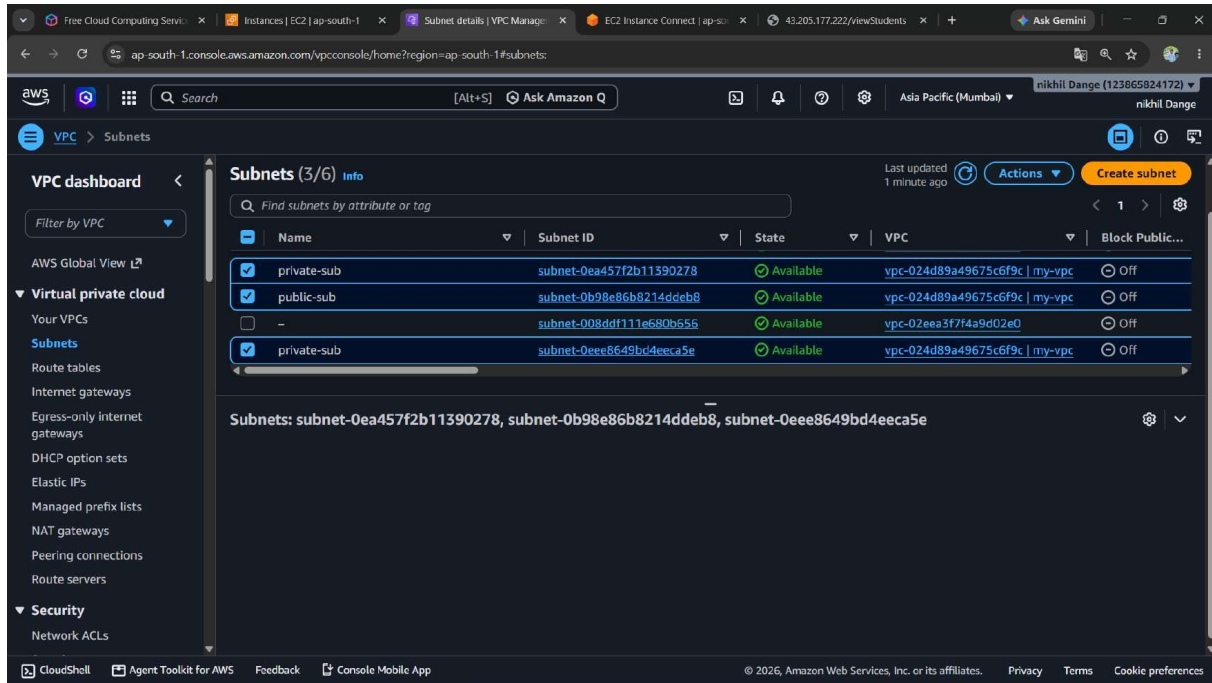


Step 2 — Subnets (Public and Private)

What it shows: The three project subnets that belong to my-vpc.

Details visible in the screenshot:

- public-sub — subnet-0b98e86b8214ddeb8
- private-sub — subnet-0ea457f2b11390278
- private-sub — subnet-0eee8649bd4eeca5e
- All three: State Available; Block Public Access: Off
- Subnet CIDRs and Availability Zones: Not visible in the provided screenshot



Step 3 — Internet Gateway

What it shows: The Internet Gateway attached to the project VPC. It gives the VPC a path to the public internet when routing and public addressing allow it.

Details visible in the screenshot:

- Name: **my-igw**; ID: igw-048dff54bcbc4334b
- State: Attached
- Attached to: my-vpc / vpc-024d89a49675c6f9c

The screenshot displays the AWS Management Console interface for the 'Internet gateways' section. The left-hand navigation pane shows the 'VPC dashboard' with a sidebar menu including 'Virtual private cloud' and 'Security'. The main content area is titled 'Internet gateways (1/2)' and contains a table listing the gateways. The gateway 'my-igw' is selected, and its details are shown below the table. The details panel indicates that the gateway is 'Attached' to the VPC 'my-vpc'.

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-0325b1638a2028f66	Attached	vpc-02eea3f7f4a9d02e0	12
my-igw	igw-048dff54bcbc4334b	Attached	vpc-024d89a49675c6f9c my-vpc	12

igw-048dff54bcbc4334b / my-igw

Details

Internet gateway ID	State	VPC ID	Owner
igw-048dff54bcbc4334b	Attached	vpc-024d89a49675c6f9c my-vpc	123865824172

Step 4 — NAT Gateway

What it shows: A public NAT Gateway placed in the public subnet. It lets private resources reach the internet for outbound traffic without becoming public themselves.

Details visible in the screenshot:

- Name: **my-nat**; ID: nat-0b52f867fc721a37a
- Connectivity type: Public; State: Available; Availability: Zonal
- Primary public IPv4: 3.7.0.89; Primary private IPv4: 10.0.3.242
- Subnet: subnet-0b98e86b8214ddeb8 / public-sub
- Created: Monday, August 3, 2026, 22:50:17 GMT+5:30

The screenshot displays the AWS Management Console interface for a NAT Gateway. The top navigation bar shows the user is logged in as 'nikhil Dange' in the 'Asia Pacific (Mumbai)' region. The left sidebar contains the 'VPC dashboard' with options like 'Filter by VPC', 'AWS Global View', and 'Virtual private cloud'. The main content area is titled 'NAT gateways (1/1)' and shows a table with one entry: 'my-nat' with ID 'nat-0b52f867fc721a37a', connectivity type 'Public', state 'Available', and availability 'Zonal'. Below the table, the 'Details' section for 'nat-0b52f867fc721a37a / my-nat' is shown, containing the following information:

Details	Connectivity type	State	State message
NAT gateway ID nat-0b52f867fc721a37a	Public	Available	-
NAT gateway ARN arn:aws:ec2:ap-south-1:123865824172:natgateway/nat-0b52f867fc721a37a	Primary public IPv4 address 3.7.0.89	Primary private IPv4 address 10.0.3.242	Primary network interface ID eni-007f7cfe53bf748d25
VPC vpc-024d89a49675c6f9c / my-vpc	Subnet subnet-0b98e86b8214ddeb8 / public-sub	Created Monday, August 3, 2026 at 22:50:17 GMT+5:30	Deleted -

Step 5 — Route Tables

What it shows: The route tables created for the project VPC.

Details visible in the screenshot:

- mt-rt (rtb-01e6a1747736b203b) — 2 explicit subnet associations; not the main table
- my-rt (rtb-01d659d6694b8875a) — associated with subnet-0b98e86b8214dd... (public-sub); not the main table
- Main route table of my-vpc: rtb-09ffbce613afd62bb
- The individual routes (targets such as the Internet Gateway or NAT Gateway): Not visible in the provided screenshot

The screenshot displays the AWS Management Console interface for the 'Route tables' section of a VPC. The left sidebar shows the navigation menu with 'Virtual private cloud' expanded, highlighting 'Route tables'. The main content area shows a table of route tables with the following data:

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
mt-rt	rtb-01e6a1747736b203b	2 subnets	—	No	vpc-024cd8
my-rt	rtb-01d659d6694b8873a	subnet-0b98e86b8214dd...	—	No	vpc-024cd8
-	rtb-09ffbce613afd62bb	—	—	Yes	vpc-024cd8
-	rtb-0fe84d19f69ceac77	—	—	Yes	vpc-02eea

Below the table, a section titled 'Route tables: rtb-01e6a1747736b203b, rtb-01d659d6694b8873a' is visible, which is currently collapsed. The top right of the console shows the user 'nikhil Dange' and the region 'Asia Pacific (Mumbai)'.

Step 6 — Security Group Rules

What it shows: The inbound rules of the security group that controls network access for the EC2 resources.

Details visible in the screenshot:

- Security group: **default**; ID: sg-0141286998e18a187; VPC: my-vpc
- 5 inbound rules: SSH (TCP 22), Custom TCP (8080), MYSQL/Aurora (TCP 3306), All traffic, HTTP (TCP 80)
- Outbound rules: 1 permission entry
- Source CIDRs / source groups of each rule: Not visible in the provided screenshot

The screenshot displays the AWS Management Console interface for a security group named 'default' (ID: sg-0141286998e18a187). The console is in the 'ap-south-1' region. The left sidebar shows the navigation menu with categories like Instances, Images, and Elastic Block Store. The main content area is titled 'Details' and shows the security group's name, ID, description, VPC ID, owner, and rule counts. Below this, there are tabs for 'Inbound rules', 'Outbound rules', 'Sharing', 'VPC associations', 'Related resources - new', and 'Tags'. The 'Inbound rules' tab is selected, showing a table with 5 rules. The table columns are Name, Security group rule ID, IP version, Type, Protocol, and Port range. The rules listed are SSH (TCP 22), Custom TCP (8080), MYSQL/Aurora (TCP 3306), All traffic (All), and HTTP (TCP 80). The 'Outbound rules' tab shows 1 permission entry.

Name	Security group rule ID	IP version	Type	Protocol	Port range
SSH	sgr-05bdbcfc83dffbcb8	IPv4	SSH	TCP	22
Custom TCP	sgr-0d067f732e634d3c0	IPv4	Custom TCP	TCP	8080
MYSQL/Aurora	sgr-0123f27cdc427295	IPv4	MYSQL/Aurora	TCP	3306
All traffic	sgr-0615418cff550a314	-	All traffic	All	All
HTTP	sgr-0b17bf4437f4e6d6c	IPv4	HTTP	TCP	80

Part B — EC2 Servers and Web Software

Step 7 — EC2 Instances

What it shows: The three EC2 instances of the project, all in the running state.

Details visible in the screenshot:

- **db-server** — i-01dd8ba0efb071790 — t3.micro — ap-south-1c
- **application-server** — i-0ede5d59eaa1cf6bb — t3.micro — ap-south-1b
- **jump-server** — i-004ccedf017a1702a — t3.micro — ap-south-1a
- Instance state: Running (status checks were still Initializing when the screenshot was taken)

The screenshot displays the AWS Management Console's 'Instances' page. The left sidebar shows navigation options like 'Instances', 'Images', and 'Elastic Block Store'. The main content area shows a table of three EC2 instances, all of which are in the 'Running' state. The instances are named 'db-server', 'application-server', and 'jump-server'. Each instance has a 'Status check' icon indicating that the checks are still 'Initializing'. The table also shows the instance ID, type (t3.micro), availability zone, and public IP address. Below the table, there is a section for 'Monitoring' with a 'Configure CloudWatch agent' button. The bottom of the console shows the 'CloudShell' and 'Agent Toolkit for AWS' buttons, along with the copyright notice for Amazon Web Services, Inc. or its affiliates.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv
db-server	i-01dd8ba0efb071790	Running	t3.micro	Initializing	ap-south-1c	-
application-se...	i-0ede5d59eaa1cf6bb	Running	t3.micro	Initializing	ap-south-1b	-
jump-server	i-004ccedf017a1702a	Running	t3.micro	Initializing	ap-south-1a	-

Step 8 — Nginx on the Jump Server and Access to the Private Server

What it shows: Nginx is validated and started on the jump-server. The jump-server is then used to log in to the private server 10.0.24.240.

Details visible in the screenshot:

- Host: jump-server (prompt ip-10-0-10-84; public IP 43.205.177.222, private IP 10.0.10.84)
- nginx configuration file /etc/nginx/nginx.conf: syntax is ok, test is successful
- nginx.service started (“The nginx HTTP and reverse proxy server”)
- Two SSH attempts failed with “Identity file B2-mumbai-key.pem not accessible” (the key was not in the current folder)
- After cd /home/ec2-user (where B2-mumbai-key.pem is present) the same command worked and logged in to ec2-user@10.0.24.240 (Amazon Linux 2023): then sudo -i was used

```
ssh -i B2-mumbai-key.pem ec2-user@10.0.24.240
```

A screenshot of a terminal window titled "EC2 Instance Connect | ap-south-1". The terminal shows the process of installing and configuring Nginx on an Amazon Linux instance. It starts with system updates, followed by installing Nginx via yum. The configuration file /etc/nginx/nginx.conf is shown, containing settings for worker processes, connections, and server blocks for http and https. The user then runs systemctl start nginx.service and systemctl status nginx.service, both showing successful results. Finally, the user runs curl -I http://localhost, which returns a 200 OK response from Nginx. The terminal interface includes a search bar at the top, navigation icons, and a footer with AWS CloudShell branding and copyright information.

Step 9 — Apache Tomcat on the Private Server

What it shows: Apache Tomcat 9.0.120 is started on the private server 10.0.24.240 (application tier).

Details visible in the screenshot:

- Path: /opt/apache-tomcat-9.0.120 (folders: bin, conf, lib, logs, temp, webapps, work)
- CATALINA_BASE and CATALINA_HOME: /opt/apache-tomcat-9.0.120; JRE_HOME: /usr/lib/jvm/java-9-openjdk-amd64
- Result: **Tomcat started.**

```
cd /opt/apache-tomcat-9.0.120/bin
./catalina.sh start
```

```

aws
[Alt+S]
Asia Pacific (Mumbai)
nikhil Dange (123865824172)
nikhil Dange

https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Aug 6 13:09:39 2026 from 10.0.10.84
[ec2-user@ip-10-0-24-240 ~]$ sudo -i
[root@ip-10-0-24-240 ~]# cd /opt
[root@ip-10-0-24-240 opt]# ls
apache-tomcat-9.0.120  aws
[root@ip-10-0-24-240 opt]# cd apache-tomcat-9.0.120/
[root@ip-10-0-24-240 apache-tomcat-9.0.120]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-10-0-24-240 apache-tomcat-9.0.120]# cd bin
[root@ip-10-0-24-240 bin]# ls
bootstrap.jar  ciphers.bat  configtest.bat  digest.sh  setclasspath.sh  startup.sh  tool-wrapper.sh
catalina-tasks.xml  ciphers.sh  configtest.sh  makebase.bat  shutdown.bat  tomcat-juli.jar  version.bat
catalina.bat  commons-daemon-native.tar.gz  daemon.sh  makebase.sh  shutdown.sh  tomcat-native.tar.gz  version.sh
catalina.sh  commons-daemon.jar  digest.bat  setclasspath.bat  startup.bat  tool-wrapper.bat
[root@ip-10-0-24-240 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.120
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.120
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.120/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.120/bin/bootstrap.jar:/opt/apache-tomcat-9.0.120/bin/tomcat-juli.jar
Tomcat started.
[root@ip-10-0-24-240 bin]#

```

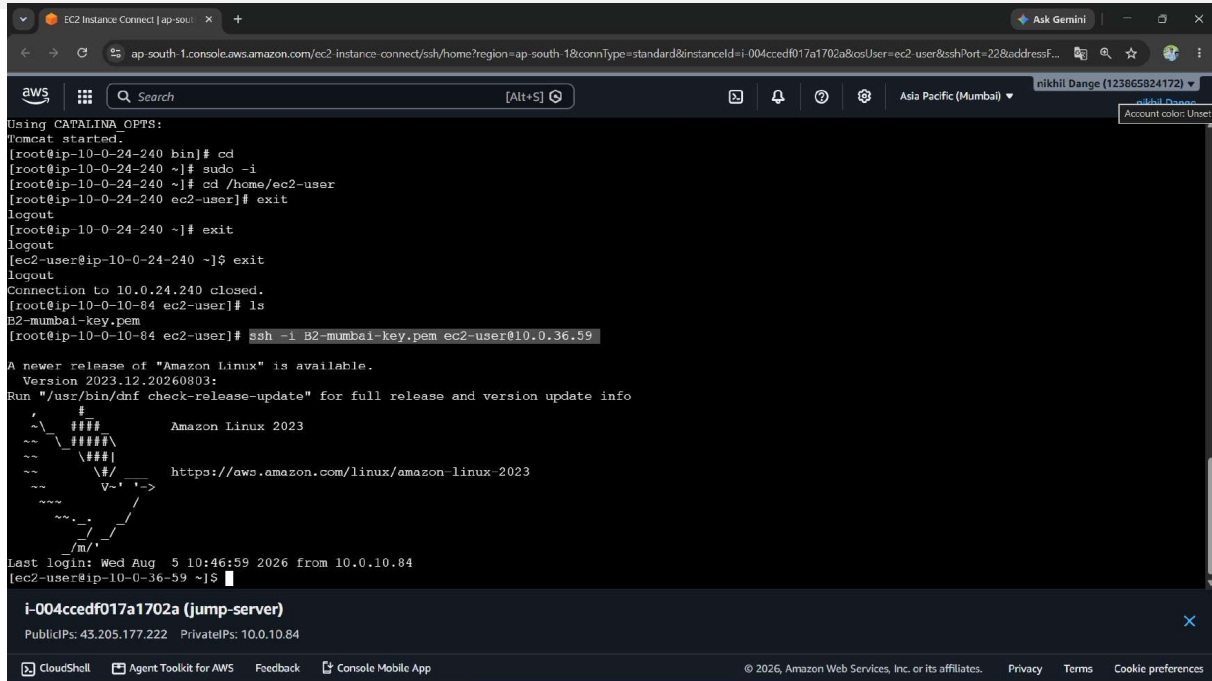
Step 10 — Access to the Second Private Server (10.0.36.59) via the Jump Server

What it shows: The session leaves 10.0.24.240, returns to the jump-server and connects to another private server that is used to reach the database.

Details visible in the screenshot:

- Connection to 10.0.24.240 closed; back on the jump-server (public 43.205.177.222, private 10.0.10.84)
- SSH to ec2-user@10.0.36.59 succeeds (Amazon Linux 2023)
- The jump-server is the only public endpoint used for reaching the private servers

```
ssh -i B2-mumbai-key.pem ec2-user@10.0.36.59
```



The screenshot shows a terminal window titled "EC2 Instance Connect [ap-sou-1]". The terminal output shows the user logging out of a previous session on 10.0.24.240 and then logging in to a new session on 10.0.36.59. The user is prompted to enter a password, and the session is successful. The terminal also displays the AWS CLI version and the user's location (Asia Pacific (Mumbai)).

```
Using CATALINA_OPTS:
[ec2-user@ip-10-0-24-240 ~]$ exit
[ec2-user@ip-10-0-24-240 ~]$ ssh -i B2-mumbai-key.pem ec2-user@10.0.36.59
Last login: Wed Aug 5 10:46:59 2026 from 10.0.10.84
[ec2-user@ip-10-0-36-59 ~]$
```

i-004ccdf017a1702a (jump-server)
PublicIPs: 43.205.177.222 PrivateIPs: 10.0.10.84

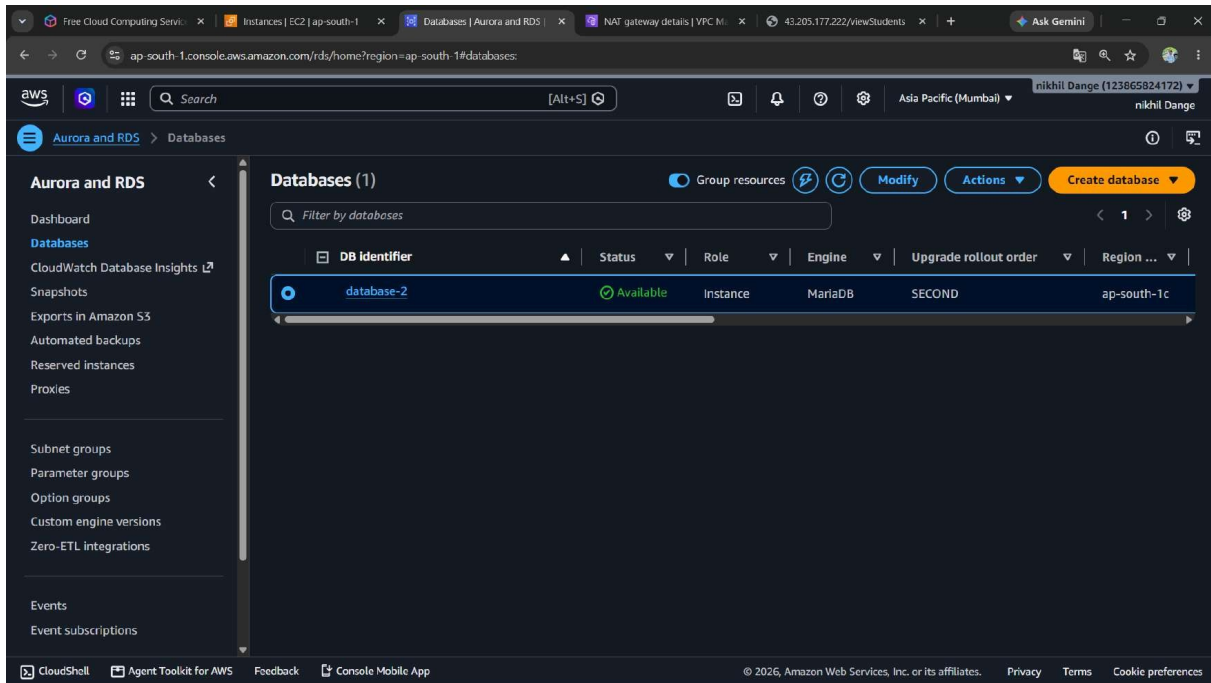
Part C — Database (Amazon RDS MariaDB)

Step 11 — RDS Database Instance

What it shows: The Amazon RDS console with the MariaDB instance used by the application.

Details visible in the screenshot:

- DB identifier: **database-2**; Engine: MariaDB; Role: Instance
- Status: Available; Upgrade rollout order: SECOND
- Region / AZ column: ap-south-1c; Databases: 1
- Instance class, storage, subnet group, public access and attached security group: Not visible in the provided screenshot



Step 12 — Download of the RDS Certificate Bundle

What it shows: The Amazon RDS global CA bundle is downloaded on 10.0.36.59 so the MariaDB connection can be verified with SSL.

Details visible in the screenshot:

- File saved as global-bundle.pem (165,408 bytes downloaded)
- Source: Amazon RDS truststore (truststore.nki.rds.amazonaws.com)

```
curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
```

[illegible]

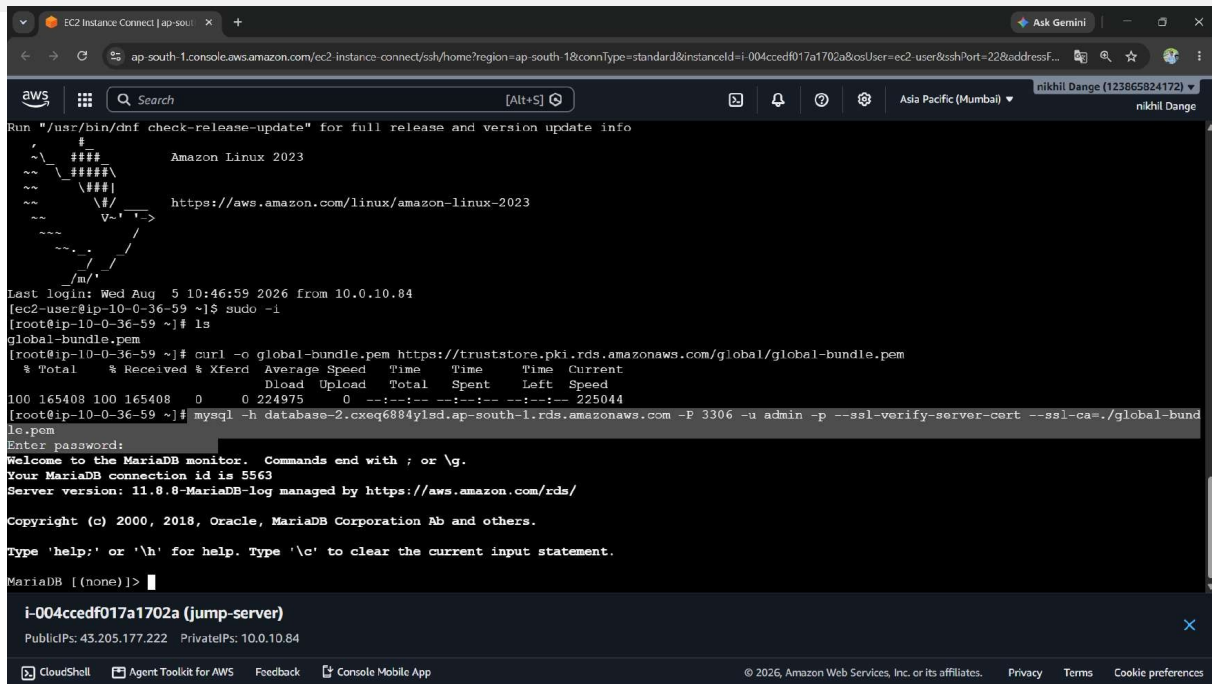
Step 13 — Connecting to MariaDB with SSL Verification

What it shows: The MariaDB client connects to the RDS endpoint using the downloaded certificate.

Details visible in the screenshot:

- Host: database-2....ap-south-1.rds.amazonaws.com; Port: 3306; User: admin
- Options: --ssl-verify-server-cert and --ssl-ca=./global-bundle.pem
- Result: “Welcome to the MariaDB monitor”; connection id 5563
- Server version: 11.8.8-MariaDB-log managed by Amazon RDS

```
mysql -h database-2.<endpoint>.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p  
--ssl-verify-server-cert --ssl-ca=./global-bundle.pem
```



The screenshot shows an AWS CloudShell terminal window. The terminal output includes the following commands and results:

```
Run "/usr/bin/dnf check-release-update" for full release and version update info
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Wed Aug 5 10:46:59 2026 from 10.0.10.84
[ec2-user@ip-10-0-36-59 ~]$ sudo -i
[root@ip-10-0-36-59 ~]# ls
global-bundle.pem
[root@ip-10-0-36-59 ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 165408 100 165408 0 0 224975 0 --:--:-- --:--:-- --:--:-- 225044
[root@ip-10-0-36-59 ~]# mysql -h database-2.cxeq6884ylsd.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 5563
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

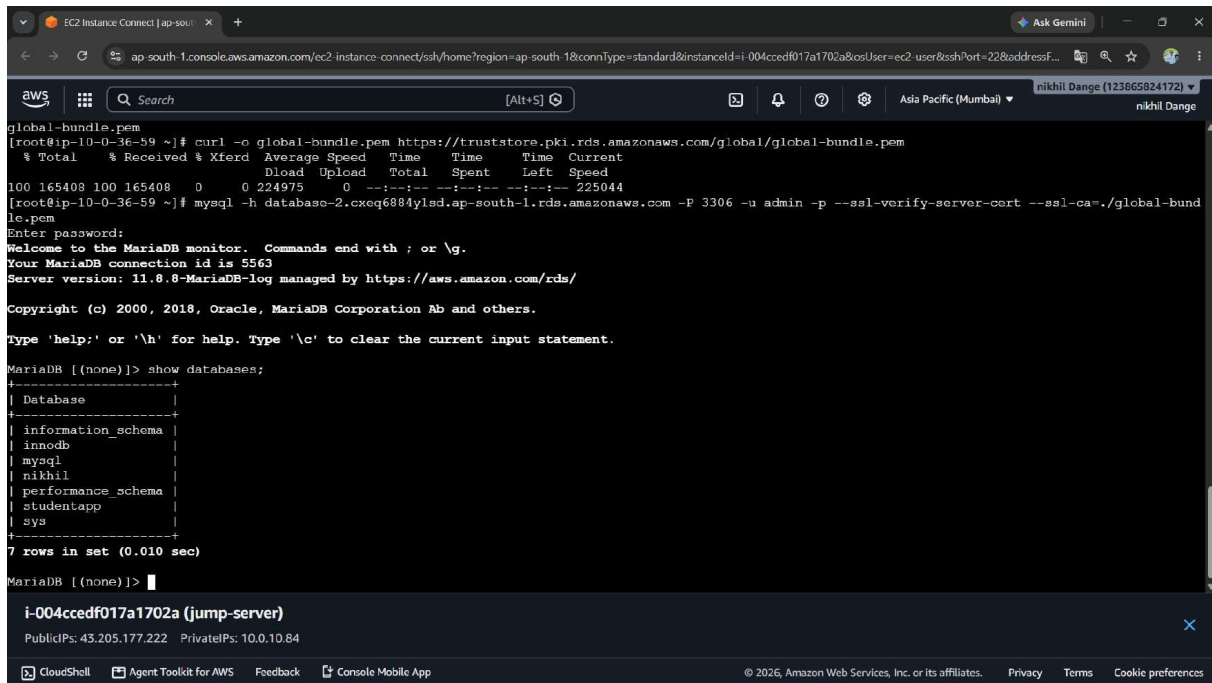
At the bottom of the terminal window, there is a summary box for the instance `i-004ccedf017a1702a (jump-server)` with Public IP `43.205.177.222` and Private IP `10.0.10.84`.

Step 14 — Listing the Databases

What it shows: The show databases; command lists what exists on the server.

Details visible in the screenshot:

- 7 rows returned: information_schema, innodb, mysql, nikhil, performance_schema, **studentapp**, sys
- The application database is **studentapp**



The screenshot shows a terminal window titled "EC2 Instance Connect [ap-sou...]" with a browser address bar showing an AWS console URL. The terminal content is as follows:

```
global-bundle.pem
[root@ip-10-0-36-59 ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total     Spent    Left     Speed
100 165408 100 165408    0     0  224975      0  --:--:-- --:--:-- --:--:-- 225044
[root@ip-10-0-36-59 ~]# mysql -h database-2.cxq6884y1sd.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bund
le.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5563
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| nikhil |
| performance_schema |
| studentapp |
| sys |
+-----+
7 rows in set (0.010 sec)

MariaDB [(none)]>
```

At the bottom of the terminal window, there is a summary bar for the instance **i-004ccedf017a1702a (jump-server)**, showing Public IP: 43.205.177.222 and Private IP: 10.0.10.84. The footer of the window includes links for CloudShell, Agent Toolkit for AWS, Feedback, Console Mobile App, and copyright information for Amazon Web Services, Inc. or its affiliates.

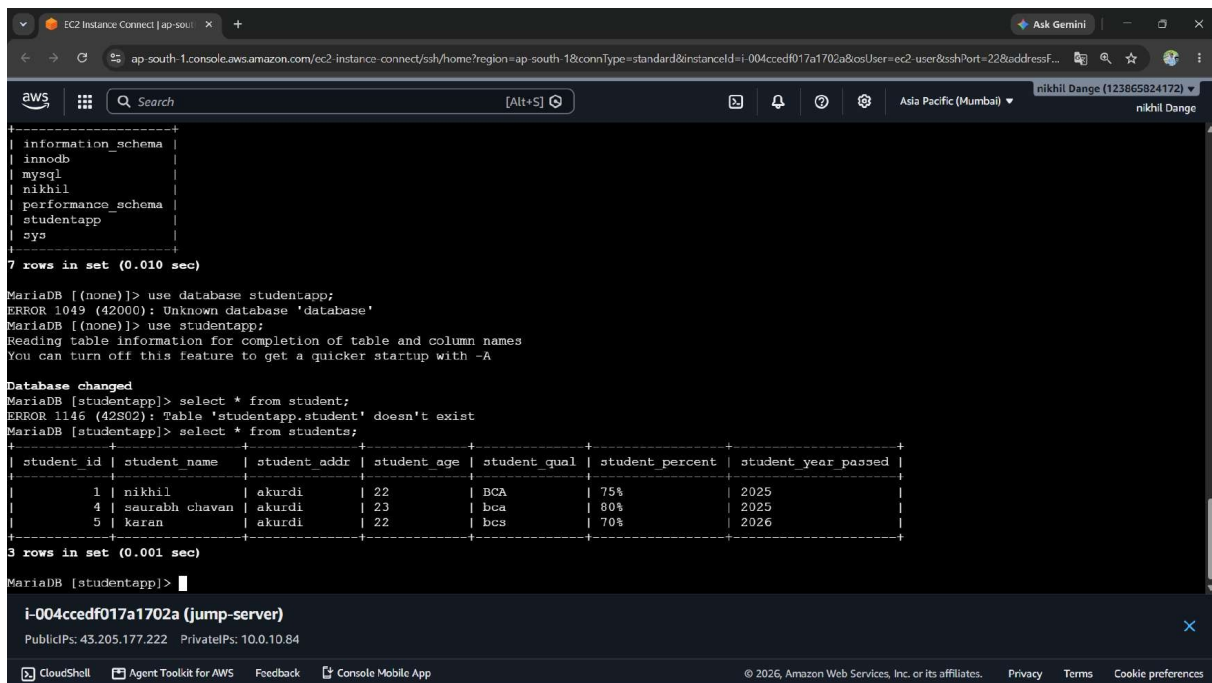
Step 15 — Selecting the Database and Querying the Student Data

What it shows: The studentapp database is selected and its data is read. Two small typing errors appear before the correct commands.

Details visible in the screenshot:

- use database studentapp; -> ERROR 1049 (Unknown database 'database')
- use studentapp; -> Database changed
- select * from student; -> ERROR 1146 (table studentapp.student doesn't exist)
- select * from students; -> 3 rows returned (table below)

student_id	student_name	student_addr	student_age	student_qual	student_percent	student_year_passed
1	nikhil	akurdi	22	BCA	75%	2025
4	saurabh chavan	akurdi	23	bca	80%	2025
5	karan	akurdi	22	bcs	70%	2026



```
aws
Search [Alt+S]
Asia Pacific (Mumbai)
nikhil Dange (123865824172)
nikhil Dange

+-----+
| information_schema |
| innodb             |
| mysql              |
| nikhil             |
| performance_schema |
| studentapp         |
| sys                |
+-----+
7 rows in set (0.010 sec)

MariaDB [(none)]> use database studentapp;
ERROR 1049 (42000): Unknown database 'database'
MariaDB [(none)]> use studentapp;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [studentapp]> select * from student;
ERROR 1146 (42S02): Table 'studentapp.student' doesn't exist
MariaDB [studentapp]> select * from students;
+-----+
| student_id | student_name | student_addr | student_age | student_qual | student_percent | student_year_passed |
+-----+
| 1          | nikhil       | akurdi       | 22          | BCA          | 75%             | 2025                |
| 4          | saurabh chavan | akurdi       | 23          | bca          | 80%             | 2025                |
| 5          | karan        | akurdi       | 22          | bcs          | 70%             | 2026                |
+-----+
3 rows in set (0.001 sec)

MariaDB [studentapp]> 
```

i-004ccdf017a1702a (jump-server)
PublicIPs: 43.205.177.222 PrivateIPs: 10.0.10.84

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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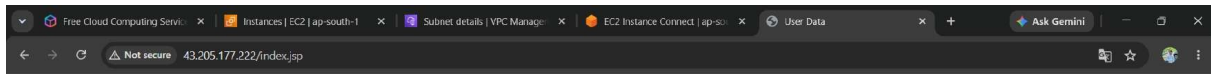
Part D — Application Output (Web Application)

Step 16 — Student Registration Form

What it shows: The web page where student details are entered.

Details visible in the screenshot:

- URL: 43.205.177.222/index.jsp (browser shows “Not secure” because the site uses HTTP)
- Fields: Student Name, Address, Age, Qualification, Percentage, Year Passed, and a register button
- Sample entry: nikhil, akurdi, 22, BCA, 75%, 2025



Student Registration Form

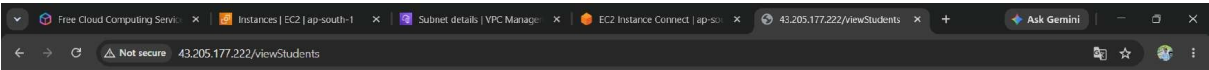
Student Name	nikhil
Student Address	akurdi
Student Age	22
Student Qualification	BCA
Student Percentage	75%
Year Passed	2025
register	

Step 17 — Students List

What it shows: The page that shows the stored students. It matches the rows found in the database.

Details visible in the screenshot:

- URL: 43.205.177.222/viewStudents
- Columns: Student ID, StudentName, Student Addr, Student Age, Student Qualification, Student Percentage, Student Year Passed, Edit, Delete
- 3 records: nikhil (ID 1), saurabh chavan (ID 4), karan (ID 5) — the same 3 rows as in Step 15



Students List

Student ID	StudentName	Student Addr	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	nikhil	akurdi	22	BCA	75%	2025	edit	delete
4	saurabh chavan	akurdi	23	bca	80%	2025	edit	delete
5	karan	akurdi	22	bcs	70%	2026	edit	delete